

Siqi Guo

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EDUCATION

Carnegie Mellon University

Master of Science in Electrical and Computer Engineering

Pittsburgh, PA

Expected May 2025

Relevant Courses: Foundations of Computer Systems, Embedded System Software Engineering, Ongoing: Parallel Computer Architecture and Programming, Distributed Systems

Tianjin University

Bachelor of Science in Computer Science and Technology; GPA: 3.84/4.0

Tianjin, China

June 2023

Relevant Courses: Operating System Principle, Data Structure, Computer Networks, Principles of Database

SKILLS

Programming Languages: C/C++, Python, C#, JavaScript, Markdown, Dart, SQL, Bash

Frameworks: NodeJS, Flask, .NET, React, Flutter, Django(Beginner), TensorFlow, PyTorch

Software and Tools: GIT, Docker, MySQL, SQLServer, Qt Creator, Visual Studio Code, Xcode, Blender(Beginner)

Platforms: MacOS, Windows, Arduino, GCP, Kaggle

INTERN EXPERIENCE

E2Biz SoftTech (Tianjin) Co. Ltd

Tianjin, China

Web Developer Intern (Full-time)

March 2023 - June 2023

- Utilized ASP.NET MVC to build the Warehousing Order Processing Unit and the Work Order Management module, designed front-end pages with Razor and KendoUI, and implemented controllers to do database filtering and searching.
- Designed an image compressor with an overall **85%** compression rate and reduced image upload time by **70%** in the project, Product Line Supermarket System.
- Cooperated with QA testers in eight function unit tests, providing test cases which increased test coverage by 21%.
- Launched this new project, ESC5Scaffold, and implemented Department Planning and User and Role Management.
- Customized the logic of privilege setting and department transferring with the controller.
- Upgraded OurCabin System by enhancing the Bug Tracking module and the FAQ section, and especially adding the Excel exporting function for all forms with **Aspose**.
- Proposed and built an Improving Item Management and Item Stage Planning subsystem with SQLServer and ASP.NET in OurCabin.
- Added the Auto-completion of historical records with **HTTP cookies and sessions** in the searching system which decreased the searching cache memory burden by **approximately 90%**.

PROJECTS

Flutter ToDo App | Personal Project 📱

Jan 2023 - Mar 2023

- Designed UI to facilitate user interactions. Executed the integration of the Firebase Authorization module, enabling secure user authentication. Developed CRUD modules for ToDo task control and utilized GCP for data storage.
- Performed integration testing and mocking with Flutter Driver after deploying this simple app.

Cache Simulator with L1 and L2 level | Tianjin University Course Project 📱

Oct 2022 - Dec 2022

- Implemented a simple Cache Simulator with L1 level Cache, applied various cache eviction strategies such as LRU, LFU, WBWA, and WTNA, and visualized the outcomes.
- Developed an advanced L2 Cache with an algorithm to effectively choose a victim cache entry.

Full-Stack Portfolio - Personal Websites | Personal Project 📱

Mar 2022 - May 2022

- Customized the schemas in the backend. Leveraged Sanity, a React-based CMS to define content models and transform contents to the front end through API integrations.
- Developed the Web architecture with React components and Animations with Framer Motion. Styled and refined adaptive UI with SCSS. Used GraphQL and GROQ to fetch dynamic data from the CMS. Finally, deployed this website on Netlify for optimal accessibility.

Bilibili Database Analyst Web App | Tianjin University Course Project 📱

Sep 2021 - Nov 2021

- Executed web crawling on the data of the Bilibili website using BeautifulSoup and managed advanced data persistence utilizing a self-designed relational MySQL database.
- Built a web-based database application using Flask, employing a client-server architecture. In addition to rendering the retrieved data, integrated CRUD functionality as well to ensure seamless data management.

Simulation of TCP Protocol | Tianjin University Course Project

Aug 2021 - Oct 2021

- Analyzed and simulated the three-way handshake protocol used by TCP with C++, conducted program refactoring and achieved a 60% reduction in code while optimizing structure, and synthesized simulation results into a report.
- Verified the results of my TCP implementation by testing the packet loss rate, simulation transmission, etc.